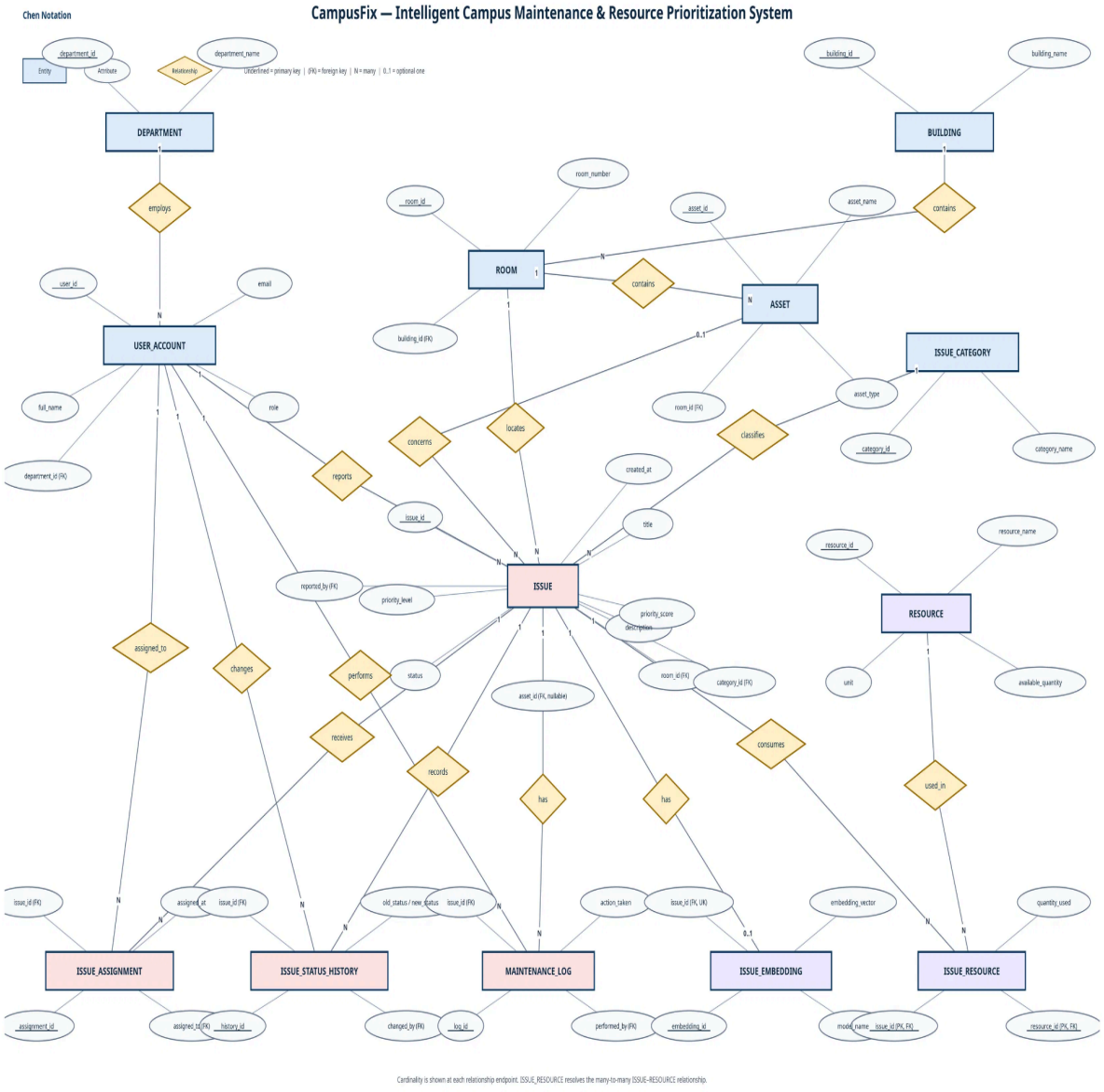
DATABASE AND MANAGEMENT SYSTEM MINI PROJECT

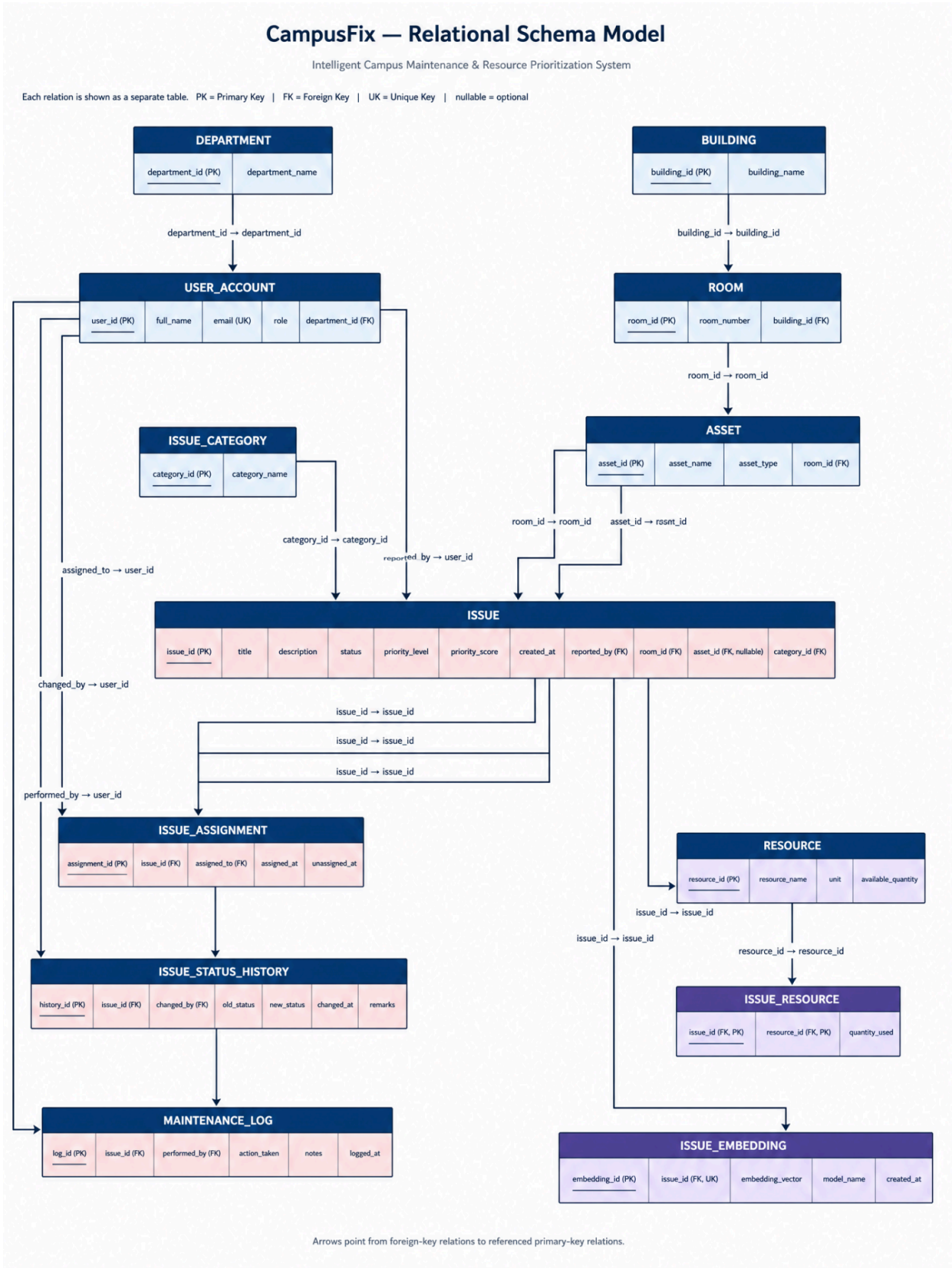
Campus Fix–Intelligence Campus Maintenance & Resource Prioritisation System

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1.E R DIAGRAM



2.RELATIONAL SCHEMA



3.DDL
SQL CODE

```sql
CREATE TABLE departments (
    department_id BIGSERIAL PRIMARY KEY,
    department_name VARCHAR(100) NOT NULL UNIQUE,
    description TEXT,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE users (
    user_id BIGSERIAL PRIMARY KEY,
    department_id BIGINT,

    full_name VARCHAR(120) NOT NULL,
    email VARCHAR(150) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    role user_role NOT NULL DEFAULT 'STUDENT',
    phone VARCHAR(20),

    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT fk_users_department
        FOREIGN KEY (department_id)
        REFERENCES departments(department_id)
        ON DELETE SET NULL
        ON UPDATE CASCADE
);

CREATE TABLE buildings (
    building_id BIGSERIAL PRIMARY KEY,

    building_name VARCHAR(120) NOT NULL UNIQUE,
    location VARCHAR(255),
    description TEXT,

    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);
```

```sql
CREATE TABLE rooms (
    room_id BIGSERIAL PRIMARY KEY,

    building_id BIGINT NOT NULL,
    room_number VARCHAR(30) NOT NULL,
    room_type VARCHAR(50),
    floor_number INTEGER,
    capacity INTEGER,

    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT fk_rooms_building
        FOREIGN KEY (building_id)
        REFERENCES buildings(building_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,

    CONSTRAINT uq_room_in_building
        UNIQUE (building_id, room_number),

    CONSTRAINT chk_room_floor
        CHECK (floor_number IS NULL OR floor_number >= 0),

    CONSTRAINT chk_room_capacity
        CHECK (capacity IS NULL OR capacity > 0)
);

CREATE TABLE issue_categories (
    category_id BIGSERIAL PRIMARY KEY,

    category_name VARCHAR(80) NOT NULL UNIQUE,
    default_severity issue_severity NOT NULL DEFAULT 'MEDIUM',
    description TEXT
);

CREATE TABLE assets (
    asset_id BIGSERIAL PRIMARY KEY,

    room_id BIGINT,
```

```
    asset_name VARCHAR(120) NOT NULL,
    asset_type VARCHAR(80) NOT NULL,
    serial_number VARCHAR(100) UNIQUE,
    purchase_date DATE,

    status asset_status NOT NULL DEFAULT 'ACTIVE',

    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT fk_assets_room
        FOREIGN KEY (room_id)
        REFERENCES rooms(room_id)
        ON DELETE SET NULL
        ON UPDATE CASCADE
);

CREATE TABLE issues (
    issue_id BIGSERIAL PRIMARY KEY,

    reported_by BIGINT NOT NULL,
    room_id BIGINT NOT NULL,
    asset_id BIGINT,
    category_id BIGINT NOT NULL,

    title VARCHAR(200) NOT NULL,
    description TEXT NOT NULL,

    severity issue_severity NOT NULL DEFAULT 'MEDIUM',
    priority_score NUMERIC(8,2) NOT NULL DEFAULT 0,

    status issue_status NOT NULL DEFAULT 'REPORTED',

    affected_users INTEGER NOT NULL DEFAULT 1,
    location_criticality INTEGER NOT NULL DEFAULT 1,
    recurrence_factor INTEGER NOT NULL DEFAULT 1,

    reported_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    assessed_at TIMESTAMP,
    resolved_at TIMESTAMP,
    closed_at TIMESTAMP,
```

```
    CONSTRAINT fk_issues_reporter
        FOREIGN KEY (reported_by)
        REFERENCES users(user_id)
        ON DELETE RESTRICT
        ON UPDATE CASCADE,

    CONSTRAINT fk_issues_room
        FOREIGN KEY (room_id)
        REFERENCES rooms(room_id)
        ON DELETE RESTRICT
        ON UPDATE CASCADE,

    CONSTRAINT fk_issues_asset
        FOREIGN KEY (asset_id)
        REFERENCES assets(asset_id)
        ON DELETE SET NULL
        ON UPDATE CASCADE,

    CONSTRAINT fk_issues_category
        FOREIGN KEY (category_id)
        REFERENCES issue_categories(category_id)
        ON DELETE RESTRICT
        ON UPDATE CASCADE,

    CONSTRAINT chk_priority_score
        CHECK (priority_score >= 0),

    CONSTRAINT chk_affected_users
        CHECK (affected_users >= 1),

    CONSTRAINT chk_location_criticality
        CHECK (location_criticality BETWEEN 1 AND 5),

    CONSTRAINT chk_recurrence_factor
        CHECK (recurrence_factor BETWEEN 1 AND 3),

    CONSTRAINT chk_resolved_date
        CHECK (
            resolved_at IS NULL
```

```
            OR resolved_at >= reported_at
        ),

    CONSTRAINT chk_closed_date
        CHECK (
            closed_at IS NULL
            OR resolved_at IS NOT NULL
        )
);

CREATE TABLE issue_assignments (
    assignment_id BIGSERIAL PRIMARY KEY,

    issue_id BIGINT NOT NULL,
    technician_id BIGINT NOT NULL,

    assigned_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    accepted_at TIMESTAMP,
    completed_at TIMESTAMP,

    assignment_notes TEXT,

    CONSTRAINT fk_assignment_issue
        FOREIGN KEY (issue_id)
        REFERENCES issues(issue_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,

    CONSTRAINT fk_assignment_technician
        FOREIGN KEY (technician_id)
        REFERENCES users(user_id)
        ON DELETE RESTRICT
        ON UPDATE CASCADE,

    CONSTRAINT uq_issue_technician
        UNIQUE (issue_id, technician_id),

    CONSTRAINT chk_assignment_dates
        CHECK (
            completed_at IS NULL
```

```
            OR completed_at >= assigned_at
        )
);

CREATE TABLE status_history (
    history_id BIGSERIAL PRIMARY KEY,

    issue_id BIGINT NOT NULL,

    old_status issue_status,
    new_status issue_status NOT NULL,

    changed_by BIGINT NOT NULL,
    changed_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

    remarks TEXT,

    CONSTRAINT fk_history_issue
        FOREIGN KEY (issue_id)
        REFERENCES issues(issue_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,

    CONSTRAINT fk_history_user
        FOREIGN KEY (changed_by)
        REFERENCES users(user_id)
        ON DELETE RESTRICT
        ON UPDATE CASCADE
);

CREATE TABLE maintenance_logs (
    log_id BIGSERIAL PRIMARY KEY,

    issue_id BIGINT NOT NULL,
    technician_id BIGINT NOT NULL,

    action_taken TEXT NOT NULL,
    remarks TEXT,

    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
```

```
    CONSTRAINT fk_maintenance_issue
        FOREIGN KEY (issue_id)
        REFERENCES issues(issue_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,

    CONSTRAINT fk_maintenance_technician
        FOREIGN KEY (technician_id)
        REFERENCES users(user_id)
        ON DELETE RESTRICT
        ON UPDATE CASCADE
);

CREATE TABLE resources (
    resource_id BIGSERIAL PRIMARY KEY,

    resource_name VARCHAR(120) NOT NULL UNIQUE,
    quantity_available NUMERIC(10,2) NOT NULL DEFAULT 0,
    unit VARCHAR(30) NOT NULL,

    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT chk_resource_quantity
        CHECK (quantity_available >= 0)
);

CREATE TABLE issue_resources (
    issue_id BIGINT NOT NULL,
    resource_id BIGINT NOT NULL,

    quantity_used NUMERIC(10,2) NOT NULL,

    PRIMARY KEY (issue_id, resource_id),

    CONSTRAINT fk_issue_resources_issue
        FOREIGN KEY (issue_id)
        REFERENCES issues(issue_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
```

```
    CONSTRAINT fk_issue_resources_resource
        FOREIGN KEY (resource_id)
        REFERENCES resources(resource_id)
        ON DELETE RESTRICT
        ON UPDATE CASCADE,

    CONSTRAINT chk_quantity_used
        CHECK (quantity_used > 0)
);

CREATE INDEX idx_users_role
    ON users(role);

CREATE INDEX idx_users_department
    ON users(department_id);

CREATE INDEX idx_rooms_building
    ON rooms(building_id);

CREATE INDEX idx_assets_room
    ON assets(room_id);

CREATE INDEX idx_assets_status
    ON assets(status);

CREATE INDEX idx_issues_reported_by
    ON issues(reported_by);

CREATE INDEX idx_issues_room
    ON issues(room_id);

CREATE INDEX idx_issues_category
    ON issues(category_id);

CREATE INDEX idx_issues_status
    ON issues(status);

CREATE INDEX idx_issues_severity
    ON issues(severity);
```

```sql
CREATE INDEX idx_issues_reported_at
    ON issues(reported_at);

CREATE INDEX idx_issues_priority
    ON issues(priority_score DESC);

CREATE INDEX idx_assignments_technician
    ON issue_assignments(technician_id);

CREATE INDEX idx_assignments_issue
    ON issue_assignments(issue_id);

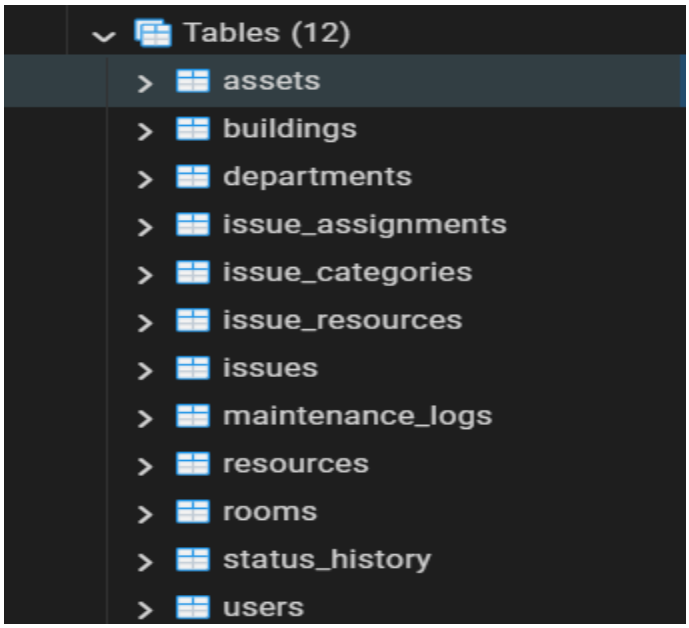
CREATE INDEX idx_status_history_issue
    ON status_history(issue_id);

CREATE INDEX idx_maintenance_issue
    ON maintenance_logs(issue_id);

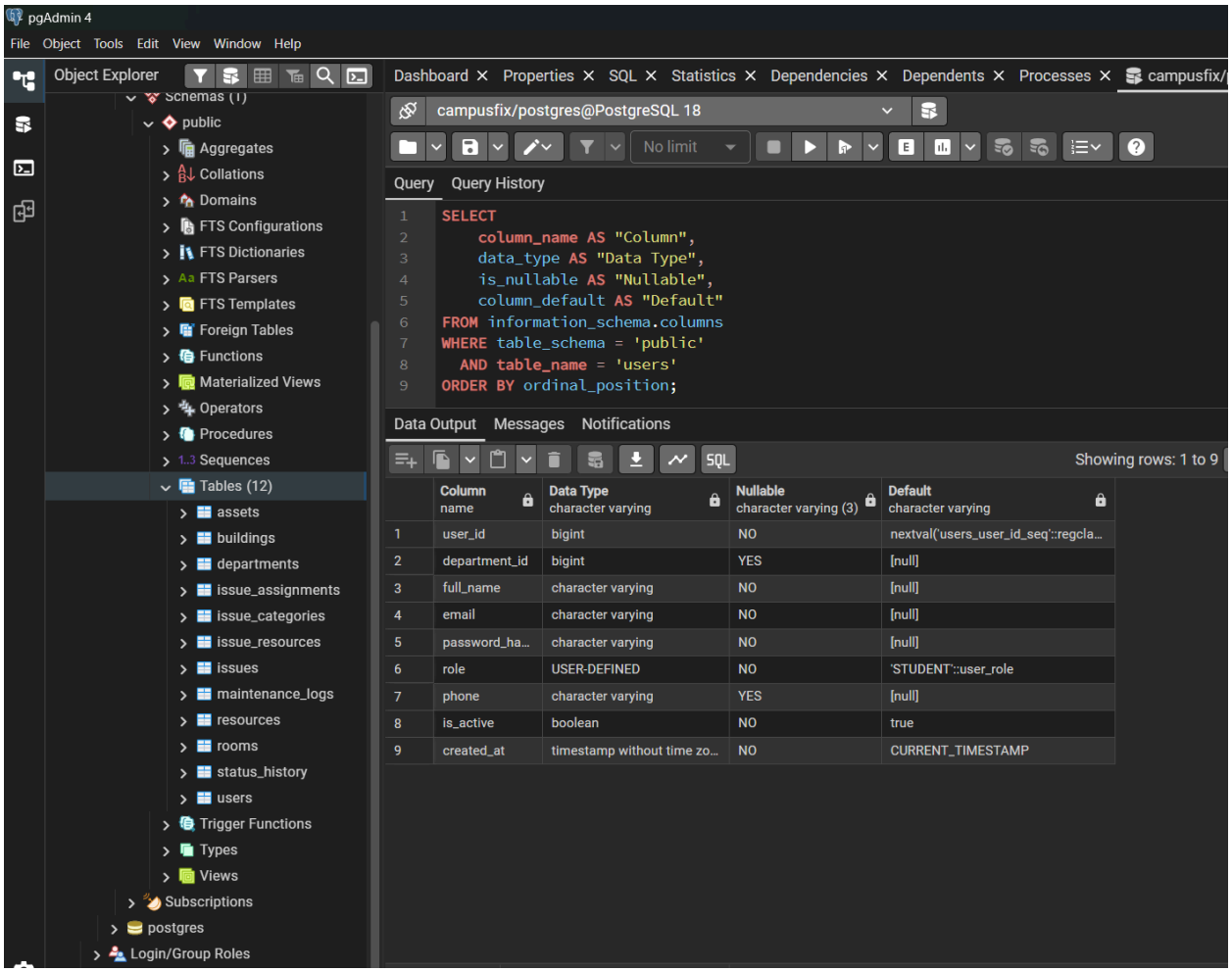
CREATE INDEX idx_maintenance_technician
    ON maintenance_logs(technician_id);

CREATE INDEX idx_issue_resources_resource
    ON issue_resources(resource_id);
```

*Tables.



*Users



*issues

Query | Query History

```
SELECT
    column_name AS "Field",
    data_type AS "Type",
    is_nullable AS "Null",
    column_default AS "Default"
FROM information_schema.columns
WHERE table_schema = 'public'
  AND table_name = 'issues'
ORDER BY ordinal_position;
```

Data Output | Messages | Notifications

Showing rows: 1 to 17

	Field name	Type character varying	Null character varying (3)	Default character varying
1	issue_id	bigint	NO	nextval('issues_issue_id_seq'::regcla...
2	reported_by	bigint	NO	[null]
3	room_id	bigint	NO	[null]
4	asset_id	bigint	YES	[null]
5	category_id	bigint	NO	[null]
6	title	character varying	NO	[null]
7	description	text	NO	[null]
8	severity	USER-DEFINED	NO	'MEDIUM'::issue_severity
9	priority_score	numeric	NO	0
10	status	USER-DEFINED	NO	'REPORTED'::issue_status
11	affected_users	integer	NO	1
12	location_critical...	integer	NO	1
13	recurrence_factor	integer	NO	1
14	reported_at	timestamp without time zo...	NO	CURRENT_TIMESTAMP
15	assessed_at	timestamp without time zo...	YES	[null]
16	resolved_at	timestamp without time zo...	YES	[null]

*buildings

Query | Query History

```
SELECT
    column_name AS "Field",
    data_type AS "Type",
    is_nullable AS "Null",
    column_default AS "Default"
FROM information_schema.columns
WHERE table_schema = 'public'
  AND table_name = 'buildings'
ORDER BY ordinal_position;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5

	Field name	Type character varying	Null character varying (3)	Default character varying
1	building_id	bigint	NO	nextval('buildings_building_id_seq'::regcla...
2	building_na...	character varying	NO	[null]
3	location	character varying	YES	[null]
4	description	text	YES	[null]
5	created_at	timestamp without time zo...	NO	CURRENT_TIMESTAMP

*issue_assignments

Query | Query History

```
SELECT
    column_name AS "Field",
    data_type AS "Type",
    is_nullable AS "Null",
    column_default AS "Default"
FROM information_schema.columns
WHERE table_schema = 'public'
  AND table_name = 'issue_assignments'
ORDER BY ordinal_position;
```

Data Output | Messages | Notifications

Showing rows: 1 to 7 | Page N

	Field name	Type character varying	Null character varying (3)	Default character varying
1	assignment_id	bigint	NO	nextval('issue_assignments_assignment_id_seq'::regcla...
2	issue_id	bigint	NO	[null]
3	technician_id	bigint	NO	[null]
4	assigned_at	timestamp without time zo...	NO	CURRENT_TIMESTAMP
5	accepted_at	timestamp without time zo...	YES	[null]
6	completed_at	timestamp without time zo...	YES	[null]
7	assignment_not...	text	YES	[null]

*MAINTENANCE LOG

Query | Query History

```
SELECT
    column_name AS "Field",
    data_type AS "Type",
    is_nullable AS "Null",
    column_default AS "Default"
FROM information_schema.columns
WHERE table_schema = 'public'
  AND table_name = 'maintenance_logs'
ORDER BY ordinal_position;
```

Data Output | Messages | Notifications

	Field name	Type character varying	Null character varying (3)	Default character varying
1	log_id	bigint	NO	nextval('maintenance_logs_log_id_seq'::regcla...
2	issue_id	bigint	NO	[null]
3	technician_...	bigint	NO	[null]
4	action_taken	text	NO	[null]
5	remarks	text	YES	[null]
6	created_at	timestamp without time zo...	NO	CURRENT_TIMESTAMP

Department

Query | Query History

```
SELECT
    column_name AS "Field",
    data_type AS "Type",
    is_nullable AS "Null",
    column_default AS "Default"
FROM information_schema.columns
WHERE table_schema = 'public'
  AND table_name = 'departments'
ORDER BY ordinal_position;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

	Field name	Type character varying	Null character varying (3)	Default character varying
1	department_id	bigint	NO	nextval('departments_departme
2	department_na...	character varying	NO	[null]
3	description	text	YES	[null]
4	created_at	timestamp without time zo...	NO	CURRENT_TIMESTAMP

Query | Query History

```
SELECT
    column_name AS "Field",
    data_type AS "Type",
    is_nullable AS "Null",
    column_default AS "Default"
FROM information_schema.columns
WHERE table_schema = 'public'
  AND table_name = 'rooms'
ORDER BY ordinal_position;
```

Data Output | Messages | Notifications

Showing rows: 1 to 7 | Page No: 1 of 1

	Field name	Type character varying	Null character varying (3)	Default character vary
1	room_id	bigint	NO	nextval('room
2	building_id	bigint	NO	[null]
3	room_num...	character varying	NO	[null]
4	room_type	character varying	YES	[null]
5	floor_numb...	integer	YES	[null]
6	capacity	integer	YES	[null]
7	created_at	timestamp without time zo...	NO	CURRENT_TI